

DEPLOY A EC2 SERVER USING TERRAFORM

1. Open you terraform (here I'am using VScode).
2. Create a file ec2, vpc, internet gateway, route, security group subnet, variables, provider, output (.tf) and terraform.tfvars.
3. Click save then in terminal type terraform init.
4. After finish type terraform plan.
5. then last type terraform apply.
6. and you can see in terminal will show you IP , press CTRL + click right for open the website.
7. Or you can go to your AWS console and go to EC2 and you will see there already one EC2 you created earlier, click to check box and click public IP IPV4 for open the website.

```
ec2.tf
1 resource "aws_instance" "web" {
2   ami           = var.ami
3   instance_type = "t3.micro"
4   subnet_id     = aws_subnet.public_subnet.id
5   vpc_security_group_ids = [aws_security_group.web_sg.id]
6   key_name      = var.key_name
7
8   user_data = <<-EOF
9     #!/bin/bash
10    yum update -y
11    yum install -y httpd
12    systemctl start httpd
13    systemctl enable httpd
14    echo "<h1>Terraform Modular SUCCESS</h1>" > /var/www/html/index.html
15    EOF
16
17   tags = {
18     Name = "web-server"
19   }
20 }
```

```
vpc.tf
1 resource "aws_vpc" "main" {
2   cidr_block = "10.0.0.0/16"
3
4   tags = {
5     Name = "vpc-terraform"
6   }
7 }
```

```

1 igw.tf
2 resource "aws_internet_gateway" "igw" {
3     vpc_id = aws_vpc.main.id
4
5     tags = {
6         Name = "igw"
7     }
8 }

```

```

1 route.tf
2 resource "aws_route_table" "rt" {
3     vpc_id = aws_vpc.main.id
4
5     route {
6         cidr_block = "0.0.0.0/0"
7         gateway_id = aws_internet_gateway.igw.id
8     }
9 }
10 resource "aws_route_table_association" "rta" {
11     subnet_id = aws_subnet.public_subnet.id
12     route_table_id = aws_route_table.rt.id
13 }

```

```

1 security_group.tf
2 resource "aws_security_group" "web_sg" {
3     name = "allow_http_ssh"
4     vpc_id = aws_vpc.main.id
5
6     ingress {
7         from_port = 80
8         to_port = 80
9         protocol = "tcp"
10        cidr_blocks = ["0.0.0.0/0"]
11    }
12    ingress {
13        from_port = 22
14        to_port = 22
15        protocol = "tcp"
16        cidr_blocks = ["0.0.0.0/0"]
17    }
18    egress {
19        from_port = 0
20        to_port = 0
21        protocol = "-1"
22        cidr_blocks = ["0.0.0.0/0"]
23    }
24 }

```

```

1 subnet.tf
2 resource "aws_subnet" "public_subnet" {
3     vpc_id = aws_vpc.main.id
4     cidr_block = "10.0.1.0/24"
5     map_public_ip_on_launch = true
6
7     tags = {
8         Name = "public-subnet"
9     }
10 }

```

```
variables.tf
1  variable "region" {
2    |   default = "ap-southeast-1"
3  }
4  variable "ami" {
5    |   default = "ami-0c0292c4186d3d1ec"
6    |   }
7
8  variable "key_name" {
9    |   default = "terraform-key"
10  }
```

```
terraform.tfvars
1  ami = "ami-0c0292c4186d3d1ec"
2  key_name = "terraform-key"
```

```
provider.tf
1  provider "aws" {
2    |   region = var.region
3  }
```

```
output.tf
1  output "public_ip" {
2    |   value = aws_instance.web.public_ip
3  }
```

de\terraform-aws-infra\provider.tf



